

Unbiased estimation via randomization - Application to Option Pricing

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(Q1) 1.

$$\begin{aligned}
 \|Y_l - Y\|_2^2 &= E[Y_l - Y]^2 \\
 &= E[Y_l^2 - 2Y_l Y + Y^2] \\
 &= E[Y_l^2] - 2E[Y_l]E[Y] + E[Y^2] \quad \text{by independence of } Y_l \text{ and } Y \\
 &\rightarrow E[Y^2] - 2E[Y]^2 + E[Y^2] \\
 &= 2 \underbrace{(E[Y^2] - E[Y]^2)}_{\text{Var}[Y]} \\
 &= 2\text{Var}[Y] \neq 0 \quad \text{if } \text{Var}[Y] \neq 0
 \end{aligned}$$

By assumption, N is almost surely finite, i.e.

$$P(N = \infty) = 0.$$

But

$$\{N \geq 1\} \supset \{N \geq 2\} \supset \{N \geq 3\} \supset \dots$$

and $\cap_{l \geq 1} \{N \geq l\} = \{N = \infty\}$. so by continuity of probability,

$$0 = P(N = \infty) = \lim_{l \rightarrow \infty} P(N \geq l),$$

so for l big enough

$$\frac{\|Y_{l-1} - Y\|_2^2}{P(N \geq l)} \geq \frac{\text{Var}[Y]}{P(N \geq l)}.$$

But since $P(N \geq l) \rightarrow 0$ we have

$$\frac{\text{Var}[Y]}{P(N \geq l)} \rightarrow \infty$$

so

$$\sum_{l=1}^{\infty} \frac{\|Y_{l-1} - Y_l\|_2^2}{P(N \geq l)} = \infty$$

2. We want to show

$$\frac{\|Y_{l-1} - Y\|_2^2}{P(N \geq l)} < \infty. \quad (1)$$

By the strong order p assumption and the fact that we have a geometric time step, we have

$$\begin{aligned} \frac{\|Y_{l-1} - Y\|_2^2}{P(N \geq l)} &\leq \frac{\left(C \left(\frac{T}{2^{l-1}}\right)^p\right)^2}{q^l} \\ &= \underbrace{CT^{2p}2^{2p}}_{\tilde{C}} \left(\frac{1}{2^{2p}q}\right)^l. \end{aligned}$$

In order for this geometric series to converge, we need $\frac{1}{2^{2p}q} < 1$, i.e. $q > 2^{-2p}$.

We now proceed to show that $\bar{Z}$ has finite variance. By definition, $\text{Var}[\bar{Z}] = E[\bar{Z}^2] - [E\bar{Z}]^2$. But by Jensen's inequality, we have

$$E[\bar{Z}] = E[Y] \leq (E[Y^2])^{1/2} < \infty \quad \text{because } Y \in L^2(\Omega).$$

So we just need to show $E[\bar{Z}^2] < \infty$. But

$$\begin{aligned} E[\bar{Z}^2] &= \sum_{l=0}^{\infty} \frac{\bar{v}_l}{P(N \geq l)} \\ &= \sum_{l=0}^{\infty} \frac{\|Y_{l-1} - Y\|_2^2 - \|Y_l - Y\|_2^2}{P(N \geq l)} \\ &\leq \sum_{l=0}^{\infty} \frac{\|Y_{l-1} - Y\|_2^2}{P(N \geq l)} < \infty \quad \text{by 1} \end{aligned}$$

(Q2) 1.

$$\begin{aligned} E[\bar{\tau}] &= E \left[\sum_{j=1}^N \bar{t}_j \right] \\ &= E \left[\sum_{j=1}^{\infty} \bar{t}_j 1(N \geq j) \right] \\ &= \sum_{j=1}^{\infty} \bar{t}_j E[1(N \geq j)] \quad \text{by Tonelli} \\ &= \sum_{j=1}^{\infty} \bar{t}_j P(N \geq j) \end{aligned}$$

2. Since $g(\alpha F) = g(F)$,

$$\min_F \left(\sum_{j=0}^{\infty} \frac{\beta_j}{F_j} \right) \left(\sum_{j=0}^{\infty} t_j F_j \right)$$

is equivalent to solving

$$\begin{aligned} \min_F \quad & \sum_{j=0}^{\infty} \frac{\beta_j}{F_j} \\ \text{subject to} \quad & \sum_{j=0}^{\infty} t_j F_j = C \end{aligned}$$

This is a constrained optimization problem so we use lagrange multipliers:

$$\mathcal{L}(F, \lambda) = \sum_{j=0}^{\infty} \frac{\beta_j}{F_j} + \lambda \left(\sum_{j=0}^{\infty} t_j F_j - C \right)$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial F_i} &= 0 \\ \frac{-\beta_i}{F_i^2} + \lambda t_i &= 0 \\ F_i &= \frac{1}{\sqrt{\lambda}} \sqrt{\frac{\beta_i}{t_i}} \end{aligned}$$

In order that $F_0 = 1$, we need

$$\frac{1}{\sqrt{\lambda}} \sqrt{\frac{\beta_0}{t_0}} = 1$$

so $\lambda = \beta_0/t_0$, i.e.

$$F_i = \frac{\sqrt{\beta_i/t_i}}{\sqrt{\beta_0/t_0}} =: F_i^*.$$

So if F^* is feasible (as stated in the assumption of the question) we have that F^* satisfies the KKT conditions. Moreover, the problem is convex so F^* is an optimal solution.

3.

$$\begin{aligned} \|\Delta_j\|_2 &= \|Y_j - Y_{j-1}\| \\ &\leq \underbrace{\|Y_j - Y\|_2}_{C2^{-pj}} + \underbrace{\|Y - Y_{j-1}\|_2}_{C2^{-p(j-1)}} \\ &\leq (C + C2^p)2^{-pj} \end{aligned}$$

so

$$\begin{aligned} E[\Delta_j^2] &= \|\Delta_j\|_2^2 \\ &= \underbrace{(C + C2^p)^2}_{\bar{C}} 2^{-2pj} \\ &= O(2^{-2pj}) \end{aligned}$$

4. In order to get the best balance between the variance and the cost, we note that we need to consider the optimal $F_j^* = P(N \geq j)$.

We know $F_j^* \sim \sqrt{\frac{\nu_j}{t_j}} \sim \sqrt{\frac{2^{-2pj}}{2^j}} = 2^{-j(p+\frac{1}{2})}$. Since we assumed that the survival distribution is of the form $P(N \geq l) = 2^{-rl}$, we have $r^* = p + \frac{1}{2}$ is balancing the variance and the cost.

Lets the conditions for $E[\bar{\tau}]$ and $Var[\bar{Z}]$ to be finite.

$$\begin{aligned} E[\bar{\tau}] &= \sum_{j=0}^{\infty} t_j P(N \geq j) \\ &\leq \sum_{j=0}^{\infty} c2^j 2^{-rj} \\ &= \sum_{j=0}^{\infty} C2^{j(1-r)} \end{aligned}$$

In order for the series to converge, we need $j(1-r) < 0$, i.e. $r > 1$.

By the same argument as in (Q1.2), $Var[\bar{Z}] < \infty$ is equivalent to $E[\bar{Z}^2] < \infty$. But

$$\begin{aligned} E[\bar{Z}^2] &= \sum_{l=0}^{\infty} \frac{\bar{\nu}_l}{P(N \geq l)} \\ &= \sum_{l=0}^{\infty} \frac{C2^{-2pl}}{2^{-rl}} \\ &= \sum_{l=0}^{\infty} C2^{l(r-2p)} \end{aligned}$$

and in order for the series to converge, we need $j(r-2p) < 0$, i.e. $r < 2p$.